The Burning Platform

Our credit portfolio earns a 2.1% true economic return and a single product category destroys $450K annually

2.1%
True Economic Return
(after charge-off losses)

$28.2M
Permanently Destroyed
(Unrecovered Losses)

13.82%
Bad Loan Rate
(Portfolio Average)

Portfolio Economics

Total Applications	38,576
Total Funded	$435.7M
Total Received	$473.1M
Gross Net Return	+$37.3M
Unrecovered Loss	–$28.2M
True Econ Return	~$9.1M
True Return Rate	2.1%
Bad Loans	5,333
Recovery Rate	56.90%

Source: financial_loan.csv | 38,576 records | $435.7M portfolio

METHODOLOGY

Analytical Approach & Data Foundation

38,576 records · $435.7M funded · 24 variables · 9 analytical dimensions · financial_loan.csv

SQL Server

Data ingestion, 9 analytical KPI queries, derived metric creation (LTI Ratio, Concentration Score), cross-validation layer for all DAX measures

Power BI

3-page interactive dashboard: Executive Command Centre | Risk Diagnostic Engine | Profitability Lens. 7 custom DAX measures.

Excel

Cross-validation of all SQL outputs, pivot table verification, scenario stress-testing, final presentation data tables

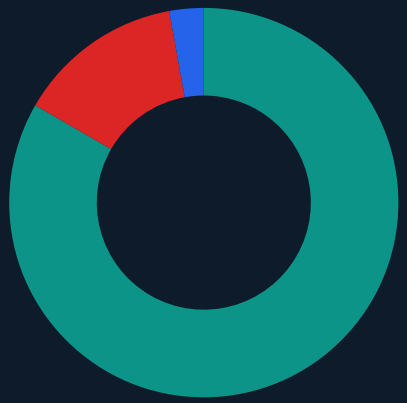
The 9 Analytical Lenses

1. Grade Risk Pricing
2. DTI Escalation Curve
3. Verification Paradox
4. Small Business Crisis
5. Purpose × Term Matrix
6. Employment Length Myth
7. LTI Cliff (Derived)
8. Geographic Concentration
9. Grade × Purpose Map

$435.7M Funded · $37.3M Earned · $28.2M Permanently Destroyed

38,576	$435.7M	$473.1M	+$37.3M	13.82%	$28.2M
Total Applications	Total Funded	Total Received	Gross Net Return	Bad Loan Rate	Unrecovered Loss

Loan Status Distribution



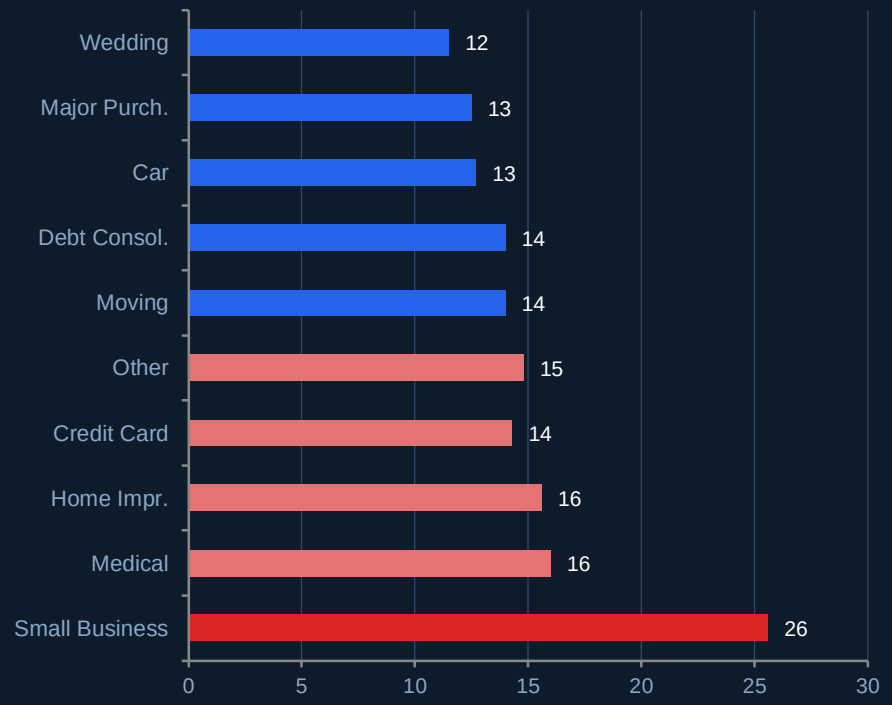
Fully Paid (83.3%) · Charged Off (13.8%) · Current (2.8%)

Default Loss Analysis

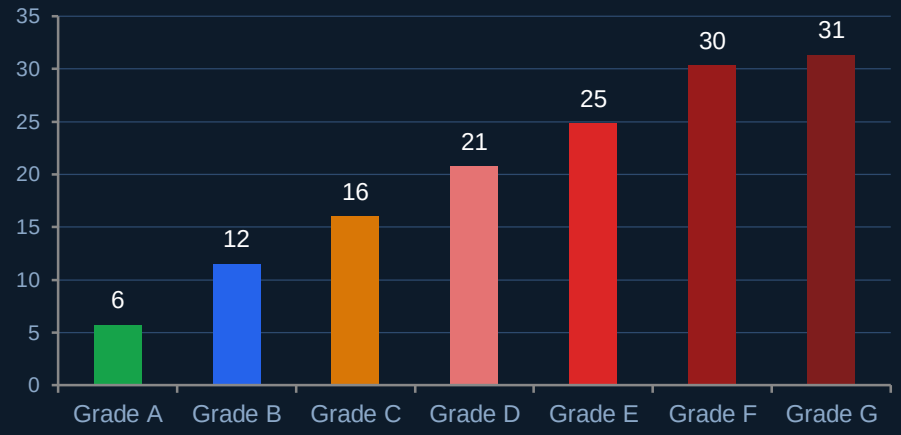
Metric	Value
Total Applications	38,576
Capital at Risk (Bad Loans)	$65,532,225
Recovered from Defaults	$37,284,763
Unrecovered True Loss	$28,247,462
Recovery Rate on Defaults	56.90%
True Economic Return Rate	~2.1%

Default Risk is Not Random — It Concentrates in Specific Segments

Default Rate % by Loan Purpose



Default Rate % by Grade



KEY FINDING

Small Biz: 25.6% default — 92% worse than portfolio average

Grade G reaches 31.3% — nearly 1 in 3 loans defaults
Portfolio average: 13.82%

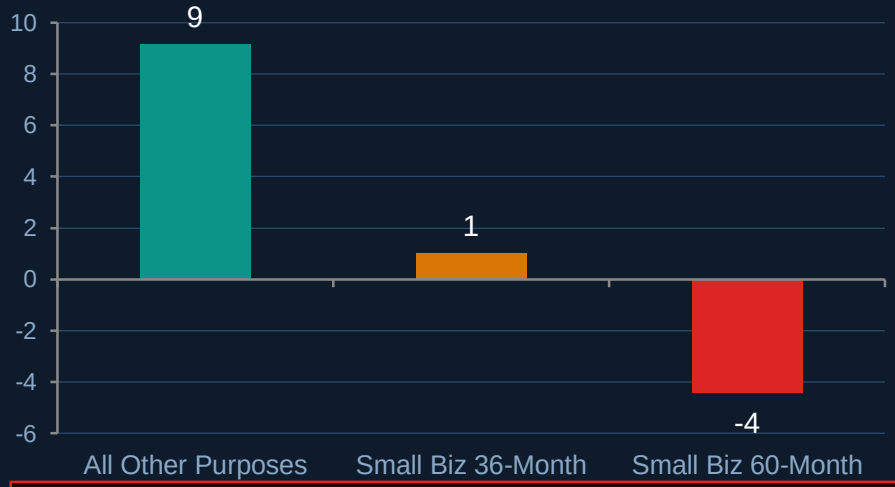
Small Business 60-Month Loans: A Confirmed Loss Product

34.55%
Default Rate
(60M Small Biz)

–$449K
Net Loss on
$10.2M Funded

25.62%
Overall Small Biz
Default Rate

Net Return % by Segment



Full Breakdown

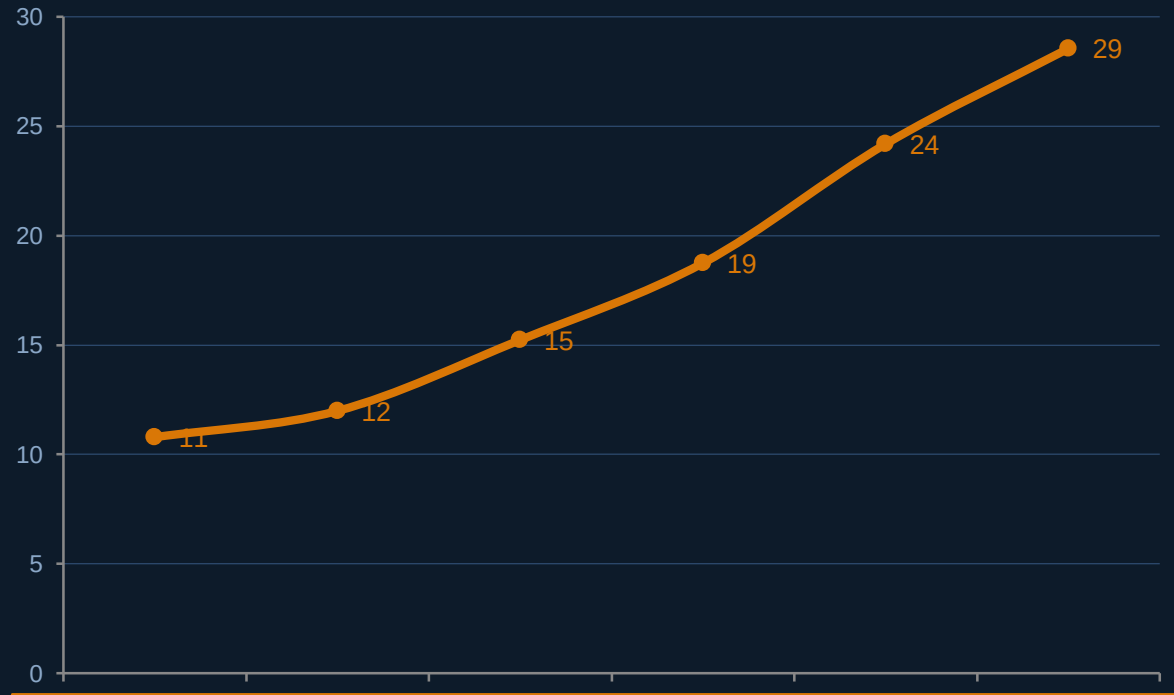
Segment	Default %	Net Ret %
All Others (36,800)	13.26%	+9.14%
All Small Biz (1,776)	25.62%	–1.28%
Small Biz 36M (1,203)	21.36%	+1.01%
Small Biz 60M (573)	**34.55%**	**–4.42%**

RECOMMENDATION: Immediately suspend 60-month small business origination → estimated +$450K annual recovery

NOVEL DERIVED INSIGHT

The Loan-to-Income Cliff: A New Risk Dimension

Borrowers taking a loan worth >30% of annual income default at 18.7% — 74% worse than sub-10% borrowers



How to Read This

LTI < 10%
10.79% default. Safe zone.

LTI 10–20%
11.99% — minimal escalation.

LTI 20–30%
15.23% — crosses portfolio avg.

LTI 30–40%
18.74% — THE CLIFF. +74% vs baseline.

LTI 40–50%
24.18% — 1 in 4 defaults. Severe.

LTI 50%+
28.54% — nearly 1 in 3 defaults.

RECOMMENDATION: Implement LTI hard cap at 35% in underwriting criteria → estimated +$2–4M annual risk reduction

MYTH-BUSTING INSIGHT

The Verification Paradox

Verified borrowers default 3.46 pp MORE — because verification tracks loan size, not creditworthiness

NOT VERIFIED

12.24%

Default Rate

Avg Loan	**$8,485**
Avg Income	**$61,607**
Avg Int Rate	**11.23%**

SOURCE VERIFIED

14.14%

Default Rate

Avg Loan	**$10,136**
Avg Income	**$68,811**
Avg Int Rate	**12.12%**

VERIFIED

15.70%

Default Rate

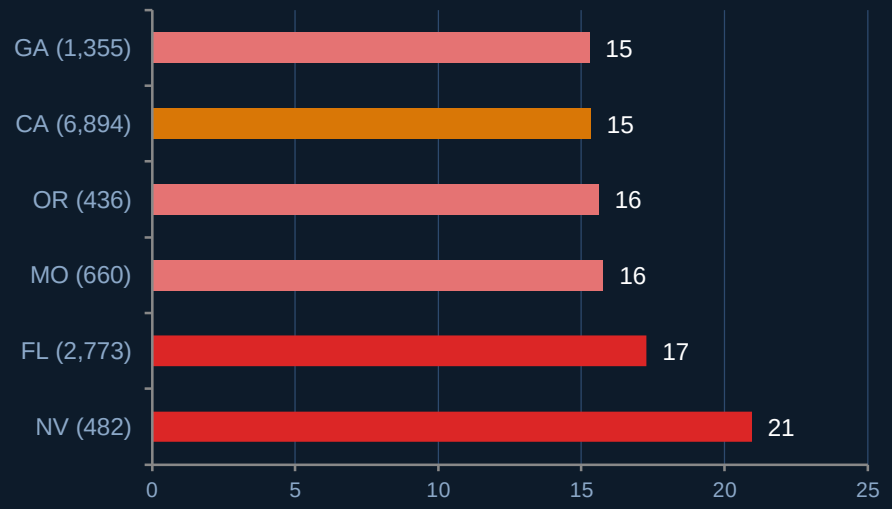
Avg Loan	**$15,968**
Avg Income	**$81,034**
Avg Int Rate	**13.09%**

EXPLANATION: Verification is triggered by loan SIZE not risk. Verified = $7,483 more borrowed on avg. Variable measures exposure, not safety.

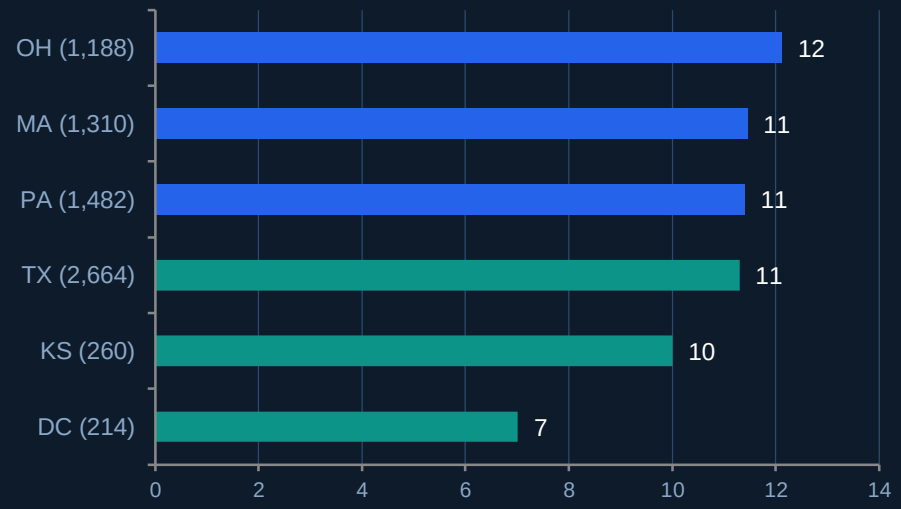
California: 18% of Portfolio at 15.33% Default — Single-State Concentration Risk

13.94 pp	18%	20.95%	7.01%
Gap: Best vs Worst State	California Portfolio Share	Nevada Default Rate (Highest)	DC Default Rate (Lowest)

Highest Default States (min. 200 loans)



Lowest Default States (min. 200 loans)



Grade × Purpose: Where the Bank is Paying to Lend

5 grade-purpose combinations with negative net returns · $396K destroyed on $16M funded

	Debt Consol.	Credit Card	Small Biz	Home Imp.	Medical	Other
Grade A	+5.2%	+5.8%	+7.2%	+6.1%	+4.5%	+5.0%
Grade B	+7.8%	+9.2%	-3.9%	+7.5%	+5.8%	+6.3%
Grade C	+9.5%	+11.0%	+6.2%	+8.8%	+7.2%	+8.1%
Grade D	+10.8%	+12.4%	-2.2%	+9.5%	+8.3%	+9.2%
Grade E	+11.2%	+13.0%	-0.8%	+10.1%	+9.0%	+9.8%
Grade F	+11.1%	+12.8%	-1.1%	+10.5%	+9.5%	+10.2%

NET RETURN %: NEGATIVE (Loss) · < 5% · 5–9% · > 9% Target

Three Policy Changes · Estimated $5–8M Annual Value Recovery

01 Suspend 60-Month Small Business Loans

IMMEDIATE

Product generates –4.42% net return with 34.55% default rate. Only confirmed loss product in the portfolio.

↑ ~+$450K annual recovery

LOW — credit policy change only, no technology.

02 Implement LTI Cap at 35%

30 DAYS

Borrowers above 35% LTI default at 18.74%+. Derived metric not in raw data — requires new origination system field.

↑ ~+$2–4M risk reduction

MEDIUM — requires LTI field in underwriting workflow.

03 Redesign Small Business Scorecard

60-90 DAYS

Small biz fails across ALL grade levels. Needs purpose-specific model: min. 2 yrs in business, collateral, co-signer.

↑ ~+$1–2M annual recovery

HIGH — new credit model and underwriting guidelines.

30–60–90 Day Implementation Plan

	Days 1–30 | Quick Wins	Days 31–60 | Policy Build	Days 61–90 | Model Redesign
Risk Team	Suspend 60M small business origination immediately	Validate LTI vs credit bureau data for accuracy	Back-test small business scorecard on historical portfolio
Credit Policy	Issue policy memo: 60M small business moratorium	Draft LTI cap policy — 35% hard limit in guidelines	Publish new small business underwriting handbook
Tech / Systems	Flag 60M SB pipeline; block new applications in system	Build LTI calculation field in origination workflow	Deploy new scorecard model in origination system

Technical Appendix — Supporting Analytics

Data Profile

38,576 rows | 24 columns | $435.7M funded | financial_loan.csv | Jan–Feb 2021 window | Zero nulls in key fields | Decimal DTI (0.20 = 20%)

SQL Validation

Master portfolio query + 9 insight queries | All KPIs cross-validated against Power BI DAX | Reproducible in MSSQL Server, PostgreSQL, or any ANSI SQL

Derived Metrics

LTI Ratio = loan_amount / annual_income | Concentration Risk Score = portfolio_share% × default_rate% | True Economic Return = Gross Return − Unrecovered Loss

Analytical Assumptions

Bad Loan = Charged Off only | Net Return = total_payment − loan_amount | Minimum 50 loans for segment credibility | DTI stored as decimal in source data

Power BI DAX Measures

Bad Loan Rate % | Net Return | Net Return % | LTI Ratio | Unrecovered Loss | Recovery Rate % | True Economic Return | 3-page dashboard

What to Investigate Next

FICO score at origination | Macro conditions at vintage | Post-charge-off collections by state | Repeat borrower behaviour | Income verification accuracy